Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 1 of 13 · Question 3 part b · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

3(a)	D; (virus)		**(1)**

Question Number	Answer	Additional guidance	Mark
3(b)	Any three from: • wear protective masks/clothes/goggles; • avoid contact with bodily fluids / avoid direct contact with infected person; • isolate/quarantine infected person; • safe burial; • disinfect surfaces to kill pathogen;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(c)	Any three from: • few pathogens enter body at infection; • large numbers required to cause symptoms; • incubation; • need to reproduce/increase numbers; • time taken to do this;	Ignore reference to virus spreading	**(3 max)**

Question Number	Answer	Additional guidance	Mark
3(d)	• symptoms include severe diarrhoea / vomiting; • loss of water/salts/dehydration; • ORT replaces water/rehydrates/replaces salts;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(e)	• causative pathogen found in bats/bats are vectors/carriers; • causes no ill effects/symptoms/disease;		**(2)**

Total 12 Marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 1 of 13 · Question 3 part b · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 2 of 13 · Question 7 part b · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

7 a)	bacteria;		1
(b)	• through water/food; • containing contaminated/infected faeces;	Do not need to say contaminated for mp2 if mentioned in mp1	2
(c)	Any four from the following: • oral rehydration therapy; • replacement of fluids/salts/sugar; • as water/salts are lost through vomiting / diarrhoea; • antibiotics; • to kill bacteria;		4

Total marks 7

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 2 of 13 · Question 7 part b · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 3 of 13 · Question 8 part a.ii · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

8 (a)	(i)	Bacteria;		(1)
	(ii)	Contaminated water/food;		(1)
(b)		• oral rehydration(therapy)/ oral rehydration (solution); • reference to electrolytes/ions/named electrolytes (mixed) in water; • Reference to clean/boiled water; • Reference to antibiotics ;		(3)

Total for Question 8 = 5 marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 4 of 13 · Question 1 part b · 4 marks

2024 JUN · Paper 02 · find part b in this mark scheme extract

Number			
1 (a)	<table><tr><th>Disease</th><th>Type of pathogen</th></tr><tr><td>cholera</td><td>Bacterium/bacteria;</td></tr><tr><td>AIDS</td><td>virus;</td></tr><tr><td>athlete's foot</td><td>Fungus/fungi;</td></tr></table>	Reject more than one line from any disease	(3)
(b)	In the following order: bacterium; sexual intercourse; barrier; antibiotics;		(4)
(c) (i)	• overall decrease; • fluctuates • steep decreases from 1990;		(2)
(ii)	17 - 9; 8;	Allow tolerance of +-0.5 ecf for mp2	(2)
(d)	**C** antigens,white blood cells produced,antibodies destroy pathogens; The following are incorrect because - A vaccinations do not contain antibodies B platelets are not produced following a vaccination D vaccinations do not produce platelets		(1)

Total for Question 1 = 12 marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 5 of 13 · Question 4 part a · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

4 (a)	• diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly; • malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ;	Ignore ORT	2
(b) (i)	antibiotics;		1
(ii)	• elderly have weakened/lowered immune system/immunocompromised; • indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together;		2
(c) (i)	• diet; • exercise; • smoking: • drinking alcohol: • stress	Accept drugs	2
(ii)	• 5 000 000 / 100 000 = 50; • 50 × 445 = 22 250; • 22 300; ***22,250 scores 2 marks*** ***22,300 scores 3 marks***	***1 mark for ×445 if there are no other marks***	3
(iii)	• use of 445 and 185; • 445 / 185 ; • 2.41:1/2.405:!/2.4:1= 3 marks;	ecf from mp1 **if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks	3
(iv)	• people live longer/elderly population; • these diseases are more common in older people; • less likely to die younger from infectious diseases/better medical facilities/treatment; • different diet/ more meat/saturated fats/processed food; • drinking more alcohol;	People live longer so there is more cancer and dementia/these diseases= 2 marks	2

Total 15 marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 5 of 13 · Question 4 part a · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 6 of 13 · Question 4 part a.ii · 2 marks

2020 JAN · Paper 02 · find part a.ii in this mark scheme extract

Number				
4	(a)(i)	pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1)	Accept fatigue	1
	(ii)	Any two from: • itching transfers eggs (from anus) to fingers/hand(1) • eggs transferred from fingers to other surfaces/people(1) • eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1)		2
	(b)	Any two from: • correct temperature/pH/provides warmth(1) • plenty of nutrients(1) • no predators(1)	Allow food	2
	(c) (i)	lacking in one or more nutrients/vitamins/minerals / other named nutrient(1)		1
	(ii)	any named nutritional deficiency such as anaemia/scurvy	Allow named nutrient e.g. protein deficiency	1
	(iii)	• pinworms absorb nutrients/named nutrient(1) • less nutrients absorbed/pass into blood/body/cells receive less nutrients(1)	Allow food	2
	(d)	Reference to eggs/pinworms on transparent tape (where they can be seen)		1
	(e)	Any two from: • refrain from itching/scratching (anus)(1) • use medication (to kill pinworms)(1) • don't share towels/clothing(1) • wash bedclothes/linen regularly/do not shake bed linen(1) • don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1) • don't allow children to bathe together(1)		2

Total for Question 4 = 12 marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 7 of 13 · Question 4 part e · 2 marks

2020 JAN · Paper 02 · find part e in this mark scheme extract

Number				
4	(a)(i)	pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1)	Accept fatigue	1
	(ii)	Any two from: • itching transfers eggs (from anus) to fingers/hand(1) • eggs transferred from fingers to other surfaces/people(1) • eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1)		2
	(b)	Any two from: • correct temperature/pH/provides warmth(1) • plenty of nutrients(1) • no predators(1)	Allow food	2
	(c) (i)	lacking in one or more nutrients/vitamins/minerals / other named nutrient(1)		1
	(ii)	any named nutritional deficiency such as anaemia/scurvy	Allow named nutrient e.g. protein deficiency	1
	(iii)	• pinworms absorb nutrients/named nutrient(1) • less nutrients absorbed/pass into blood/body/cells receive less nutrients(1)	Allow food	2
	(d)	Reference to eggs/pinworms on transparent tape (where they can be seen)		1
	(e)	Any two from: • refrain from itching/scratching (anus)(1) • use medication (to kill pinworms)(1) • don't share towels/clothing(1) • wash bedclothes/linen regularly/do not shake bed linen(1) • don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1) • don't allow children to bathe together(1)		2

Total for Question 4 = 12 marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 8 of 13 · Question 2 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
2 (a)	(i)	• genetic material made of RNA (1) • no DNA (1)		2
	(ii)	<table><tr><th>Disease</th><th>Blood Tested (✓)</th></tr><tr><td>anaemia</td><td></td></tr><tr><td>cystic fibrosis</td><td></td></tr><tr><td>gonorrhoea</td><td>✓</td></tr><tr><td>HIV</td><td>✓</td></tr><tr><td>scurvy</td><td></td></tr></table>	-1 for each extra tick	2
	(iii)	• donor/blood transfused into person with Ebola (1) • needs to be compatible/matched/same group/not rejected (1) • otherwise agglutination/clumping occurs (1)	R clotting	3
	(iv)	• cause blood cells to burst/cells are damaged/destroyed/lose structure (1) • results in loss of function (1)		2
	(v)	• antibodies in donated blood (1) • can destroy virus in infected person (1)		2
(b)		• uracil present, not thymine (1) • guanine would pair with cytosine/adenine with uracil (1) • percentage of G and C/A and U would be the same (1)		3

Total for question = 14 marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 8 of 13 · Question 2 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 9 of 13 · Question 6 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number					
6	(a)	(i)	phagocytosis;		1
		(ii)	• enclosed bacterium fuses with lysosome; • enzymes break down /digests bacterium; • remains of bacterium removed from white blood cell;		3
		(iii)	• label/identify/complementary to bacterium/antigens presented; • so response quicker if new infection/better immune response;		2
	(b)	(i)	any three from • prick skin to obtain blood; • use a sterile needle/pin • smear/spread blood on slide; • ensure smear is thin; • add stain; • allow to dry;		3
		(ii)	any four from • red blood cells would be seen as circle/not in 3D; • white blood cells would be larger than red/rbc too large; • wouldn't see platelets; • because too small for light microscope;		4
		(iii)	• no risk of infection; • e.g. AIDS;	R contaminate	2

Total 15

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 9 of 13 · Question 6 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 10 of 13 · Question 3 part e · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part e in this mark scheme extract

3(a)	D; (virus)		**(1)**

Question Number	Answer	Additional guidance	Mark
3(b)	Any three from: • wear protective masks/clothes/goggles; • avoid contact with bodily fluids / avoid direct contact with infected person; • isolate/quarantine infected person; • safe burial; • disinfect surfaces to kill pathogen;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(c)	Any three from: • few pathogens enter body at infection; • large numbers required to cause symptoms; • incubation; • need to reproduce/increase numbers; • time taken to do this;	Ignore reference to virus spreading	**(3 max)**

Question Number	Answer	Additional guidance	Mark
3(d)	• symptoms include severe diarrhoea / vomiting; • loss of water/salts/dehydration; • ORT replaces water/rehydrates/replaces salts;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(e)	• causative pathogen found in bats/bats are vectors/carriers; • causes no ill effects/symptoms/disease;		**(2)**

Total 12 Marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 10 of 13 · Question 3 part e · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part e in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 11 of 13 · Question 3 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

3(a)	D; (virus)		**(1)**

Question Number	Answer	Additional guidance	Mark
3(b)	Any three from: • wear protective masks/clothes/goggles; • avoid contact with bodily fluids / avoid direct contact with infected person; • isolate/quarantine infected person; • safe burial; • disinfect surfaces to kill pathogen;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(c)	Any three from: • few pathogens enter body at infection; • large numbers required to cause symptoms; • incubation; • need to reproduce/increase numbers; • time taken to do this;	Ignore reference to virus spreading	**(3 max)**

Question Number	Answer	Additional guidance	Mark
3(d)	• symptoms include severe diarrhoea / vomiting; • loss of water/salts/dehydration; • ORT replaces water/rehydrates/replaces salts;		**(3)**

Question Number	Answer	Additional guidance	Mark
3(e)	• causative pathogen found in bats/bats are vectors/carriers; • causes no ill effects/symptoms/disease;		**(2)**

Total 12 Marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 11 of 13 · Question 3 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 12 of 13 · Question 4 part a.i · 1 mark

2020 JAN · Paper 02 · find part a.i in this mark scheme extract

Number			
4 (a)(i)	pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1)	Accept fatigue	1
(ii)	Any two from: • itching transfers eggs (from anus) to fingers/hand(1) • eggs transferred from fingers to other surfaces/people(1) • eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1)		2
(b)	Any two from: • correct temperature/pH/provides warmth(1) • plenty of nutrients(1) • no predators(1)	Allow food	2
(c) (i)	lacking in one or more nutrients/vitamins/minerals / other named nutrient(1)		1
(ii)	any named nutritional deficiency such as anaemia/scurvy	Allow named nutrient e.g. protein deficiency	1
(iii)	• pinworms absorb nutrients/named nutrient(1) • less nutrients absorbed/pass into blood/body/cells receive less nutrients(1)	Allow food	2
(d)	Reference to eggs/pinworms on transparent tape (where they can be seen)		1
(e)	Any two from: • refrain from itching/scratching (anus)(1) • use medication (to kill pinworms)(1) • don't share towels/clothing(1) • wash bedclothes/linen regularly/do not shake bed linen(1) • don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1) • don't allow children to bathe together(1)		2

Total for Question 4 = 12 marks

Type MS · Describe how a named infection spreads, shows itself and is prevented

Answer 13 of 13 · Question 4 part d · 1 mark

2020 JAN · Paper 02 · find part d in this mark scheme extract

Number				
4	(a)(i)	pinworms/eggs in faeces/nutritional deficiency/named deficiency/weight loss/malnutrition(1)	Accept fatigue	1
	(ii)	Any two from: • itching transfers eggs (from anus) to fingers/hand(1) • eggs transferred from fingers to other surfaces/people(1) • eggs ingested when other people touch (contaminated) surfaces (and put hands in their mouth)(1)		2
	(b)	Any two from: • correct temperature/pH/provides warmth(1) • plenty of nutrients(1) • no predators(1)	Allow food	2
	(c) (i)	lacking in one or more nutrients/vitamins/minerals / other named nutrient(1)		1
	(ii)	any named nutritional deficiency such as anaemia/scurvy	Allow named nutrient e.g. protein deficiency	1
	(iii)	• pinworms absorb nutrients/named nutrient(1) • less nutrients absorbed/pass into blood/body/cells receive less nutrients(1)	Allow food	2
	(d)	Reference to eggs/pinworms on transparent tape (where they can be seen)		1
	(e)	Any two from: • refrain from itching/scratching (anus)(1) • use medication (to kill pinworms)(1) • don't share towels/clothing(1) • wash bedclothes/linen regularly/do not shake bed linen(1) • don't bite nails/keep fingernails short / clean/wash hands (after using toilet)(1) • don't allow children to bathe together(1)		2

Total for Question 4 = 12 marks